

Strange Constellations (constellation)

Back when he was little, Stefan used to look a lot at the sky during the night, hoping to see various kinds of stars and to imagine different kinds of shapes which can be drawn amongst them. Looking back at his childhood, he realized that he can now use the computer to refine this process!



Figure 1: Is that a horse?

Therefore, he gives you the 2D map of the stars during one particular night, and he tells you that there were N stars on the sky. Furthermore, he also tells you the position (x_i, y_i) of each star.

Stefan is interested in finding **parallel L -shaped constellations**. A parallel L -shaped path is a sequence of three stars which form an L -shape and each side is parallel to the coordinate axes. The L can have its sides rotated, as long as it still forms an L . Furthermore, the L can't be degenerated (i.e. a straight line or a dot).

Help Stefan find the number of parallel L -shaped constellations in the given star map!

 Among the attachments of this task you may find a template file `constellation.*` with a sample incomplete implementation.

Input

The first line contains the only integer N .

The next N line contains 2 integers x_i and y_i .

Output

You need to write a single line with an integer: the number of parallel L -shaped constellation.

Constraints

- $1 \leq N \leq 200\,000$.
- $0 \leq x_i, y_i \leq 200\,000$ for each $i = 0 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- Subtask 1 (0 points) Examples.
 
- Subtask 2 (25 points) $N \leq 200$.
 
- Subtask 3 (40 points) $N \leq 2000$.
 
- Subtask 4 (35 points) No additional limitations.
 

Examples

input	output
6 5 5 8 5 5 8 3 3 5 1 0 5	4

Explanation

In the first sample case there are 4 *L*-shaped constellation.

